

Jonathan Copeland Williams

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EDUCATION

Princeton University

Ph.D. Computer Science | Advisor: Karthik Narasimhan

Princeton, NJ

Expected 2030

Stanford University

B.S Mathematics | Advisors: Chelsea Finn, Yoonho Lee, Pranav Rajpurkar

Palo Alto, CA

2025

PUBLICATIONS

- Jonathan Williams, Esin Tureci, Karthik Narasimhan. “Adapter Thickets: Splitting an RLVR Budget Beats Concentrating It” (Under Review @ ICLR 2027, NeurIPS 2026 Workshop Pre-to-Post)
- Jonathan Williams, Esin Tureci. “Prioritize the Process, Not Just the Outcome: Rewarding Latent Thought Trajectories Improves Reasoning in Looped Language Models” (ICML 2026)
- Allison Chen, William Yang, Salma Abdel Magid, Jonathan Williams, Olga Russakovsky, Esin Tureci. “An In-Depth Analysis of Hallucination Detection Methods for Vision-Language Models” (Under Review @ NeurIPS 2026)
- Yoonho Lee, Jonathan Williams, Henrik Marklund, Archit Sharma, Eric Mitchell, Anikait Singh, Chelsea Finn. “Test-Time Alignment via Hypothesis Reweighting” (TMLR 2026, ICML 2025 Workshop PUT)

HONORS & AWARDS

- NSF Graduate Research Fellowship
- GEM Fellowship
- Princeton President’s Fellowship

WORK EXPERIENCE

IBM Research

Machine Learning Research Intern

Yorktown Heights, NY

June 2025 - Sept 2025

- Conducted Mechanistic Interpretability Analysis of an IBM on Time Series Forecasting Model. Work was contracted under an NDA.

Stanford AI Lab (SAIL) - Finn Group

Machine Learning Research Intern

Palo Alto, CA

Oct 2023 - Present

- Enhanced Test-Time algorithm (HyRe) which improves ensemble model performance Out of Distribution by weighting member predictions proportionally to accuracy. Applied HyRe to various UCI datasets. Outperformed multiple robustness baselines (IRM, CORAL, Group DRO, Fish, LISA, etc.) on the WILDS dataset Camelyon17.
- Created pipeline to describe differences between image distributions in natural language. LLaVA was used for Image Captioning. KeyBert, ResNets, and CLIP were used for identification of optimal discriminatory key-phrases. Applied pipeline to the WILDS dataset iWildCam and captured meaningful differences (ex: "lush green fields").

Harvard Medical School - Rajpurkar Lab

Machine Learning Research Intern

Boston, MA

June 2022 - Aug 2023

- Created Contrastive Models with Multilingual text encoders (XGLM-564M, Bert Multilingual Cased, XLMRoberta, LaBSE, InfoXLM, etc). Performed zero-shot Chest X-Ray disease classification on the English MIMIC-CXR and Spanish Padchest datasets.
- Created a Coordinate Regression CNN to detect proper Endotracheal Tube placement from AP-view Chest X-Ray. Collated training data by writing a regex script to parse distance annotations from MIMIC-CXR radiology reports.

Consumer Reports

Machine Learning Engineering Intern

Yonkers, NY

May 2021 - Sept 2021

- Constructed sentiment analysis NLP pipeline. Work was contracted under an NDA.

UPMC Hillman Cancer Center - Zervantonakis Lab

Mathematical Oncology Research Intern

Pittsburgh, PA

June 2020 - Nov 2020

- Created an ODE system to model the PI3K-AKT cell signaling pathway which is deregulated in many HER2+ Breast Cancers. Reinforced previously published results and elucidated 1 new means to suppress HER2+ growth.